

Numerical Computation and Stability

Course notes - Department of Mathematics

1. Floating point arithmetic

Real numbers are represented in finite precision, so almost every value is stored with a relative error bounded by the machine epsilon. Floating point addition is not associative, and comparing computed values for exact equality is a defect.

Catastrophic cancellation occurs when subtracting nearly equal quantities: the leading digits cancel and the relative error in the result is dominated by earlier rounding. Reformulating the expression usually avoids it.

2. Root finding

Bisection brackets a root and halves the interval each step. It converges linearly but is unconditionally reliable once a sign change is found.

Newton's method converges quadratically near a simple root but needs the derivative and can diverge from a poor starting point. The secant method approximates the derivative by finite differences, trading a little convergence speed for not requiring it.

3. Interpolation

A polynomial of degree n passes through any $n+1$ distinct points, and that polynomial is unique. High-degree interpolation on equally spaced nodes oscillates wildly near the interval ends, which is Runge's phenomenon.

Splines avoid this by fitting low-degree polynomials piecewise with continuity conditions at the knots, which is why cubic splines dominate in practice.

4. Numerical integration

The trapezoidal rule approximates the integrand by straight segments with error $O(h^2)$. Simpson's rule uses quadratic segments and achieves $O(h^4)$ for smooth integrands.

Gaussian quadrature chooses both nodes and weights optimally, integrating polynomials of degree $2n-1$ exactly with n points. Adaptive schemes concentrate effort where the integrand varies most.

5. Conditioning and stability

Conditioning is a property of the problem: an ill-conditioned problem amplifies input perturbations regardless of the algorithm used. Stability is a property of the algorithm: a stable algorithm does not amplify rounding beyond what conditioning already implies.

Solving a linear system whose condition number is large means losing roughly that many digits of accuracy. Recognising an ill-conditioned problem is more valuable than any refinement of the solver.